Class 10 Math - Chapter 4: Quadratic Equations

🌟 Introduction

A **quadratic equation** in the variable x is an equation of the form: $$ ax^2 + bx + c = 0 $$ where $a, b,$ and c are real numbers, and $a \neq 0$.

Quadratic equations arise in numerous real-life applications involving projectile motion, area calculations, and profit maximization.

📌 Methods of Solving Quadratic Equations

There are three primary algebraic methods to solve quadratic equations:

1. **Factorisation Method:** Splitting the middle term so that the product of the parts equals $a \times c$.
2. **Completing the Square:** Adding and subtracting a constant to form a perfect square binomial.
3. **Quadratic Formula:** Directly plugging $a, b,$ and c into the universal formula.

> [!TIP] The **Quadratic Formula** is the most reliable method and works for all quadratic equations, even when factorisation fails! $$ x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} $$

Mind Map of Solving Methods

```
graph TD;
    Quad_Equation-->Factorisation;
    Quad_Equation-->Quadratic_Formula;
```

```
    Quad_Equation-->Completing_Square;
    Factorisation-->Middle_Term_Split["Middle Term Split"];
    Quadratic_Formula-->Discriminant["Use Discriminant"];
    Completing_Square-->Perfect_Square["Create Perfect Square"];
```

⚡ Role of the Discriminant (D)

The term under the square root in the quadratic formula, $b^2 - 4ac$, is called the **Discriminant** (D). It dictates the **Nature of Roots**.

Value of Discriminant ($D = b^2 - 4ac$)	Nature of Roots	Real World Meaning
$D > 0$	Two distinct real roots	The parabola crosses the x-axis twice.
$D = 0$	Two equal real roots	The parabola touches the x-axis precisely once.
$D < 0$	No real roots	The parabola never touches the x-axis.

❌ Common Mistakes

> [!WARNING]
>
> 1. **Signs under the root:** Miscalculating $b^2 - 4ac$. For example, if a and c are both negative, then $-4ac$ evaluates to a negative number! Wrap components in parentheses when substituting.
> 2. **Neglecting $\pm$:** When taking a square root on both sides of an equation (e.g. $x^2 = 16$), the solution is $x = \pm 4$, not just 4.

> 3. **Non-Standard Form:** Never extract a, b, c before expanding the equation into standard $ax^2 + bx + c = 0$ form. For example, $(x-2)^2 = 5$ isn't ready for a, b, c extraction.

📝 Quick Revision

Example Problem

Find the nature of the roots of $2x^2 - 4x + 3 = 0$. **Solution:** Here $a = 2, b = -4, c = 3$. $D = b^2 - 4ac = (-4)^2 - 4(2)(3)$ $D = 16 - 24 = -8$ Since $D < 0$, the equation has **no real roots**.

Summary

Quadratic equations deal with squares and curves. The Discriminant provides an instant test for whether a solution is even possible, while the Quadratic Formula serves as the ultimate tool to find exact coordinates for roots. Mastery of completing the square also builds a strong foundation for future calculus and coordinate geometry topics!